

Full length article

Autonomous ferries in light of labor regulations—A passenger perspective

Christian Braathen, Julio C. Goetz, Mario Guajardo *

NHH Norwegian School of Economics, Helleveien 30, NO-5045, Bergen, Norway

ARTICLE INFO

Keywords:

Public transportation

Autonomous ferries

Integer linear programming

ABSTRACT

We study the potential effects of introducing autonomous ferries in a transportation system of water buses. We develop two integer linear programming models and a heuristic to find weekly passenger transportation plans. One model is tailored for a fleet of autonomous ferries and the other one for manually operated ferries. The objective of the models is to minimize a penalty function for unmet demand, adding up penalties on time delays with respect to the wished time of arrivals of the passengers and penalties on the assignment of passengers to alternative transportation modes. The models differ because working laws affect the crews' working capacities, and we study the changes when these requirements are absent with autonomous ferries. Our work is motivated by the case of Bergen, a coastal city in Norway. In this case, the use of autonomous ferries has the potential to improve passengers' utility significantly. However, we suggest that it may be beneficial to consider autonomous ferries as a complementary alternative that can operate especially in low-demand hours—a recommendation that may be particularly relevant if there are few autonomous ferries available or the ferries can only be operative for a limited number of hours of the day.

1. Introduction

The use of autonomous vessels (AVs) is a highly topical issue due to the increasing capabilities of artificial intelligence and changing regulations allowing their use in national and international waters. The Finnish archipelago was a significant milestone in the development of AVs with the implementation of the world's first fully autonomous ferry, the "Falco", in 2018 (Rolls-Royce Holdings plc, 2018). This implementation serves as a proof of concept for the potential use of AVs in public transportation. However, to this date, too little focus has been given to the end-users of autonomous vessels. While the development of AVs has focused heavily on the technology and the literature has focused primarily on the feasibility, implementation, and assessment of autonomous ships, there has yet to be a good understanding of the potential gains of end-users from using autonomous vessels compared to manually operated vessels.

In this article, we are interested in the question of what are the operational implications and benefits for passengers when ferries are operated autonomously instead of manually. To address this question, we develop two integer linear programming models and a heuristic, whose main goal is to match transportation demand of passengers with a fleet of ferries on a weekly plan basis. One of these models is tailored for a fleet of autonomous ferries, while the other model is tailored for manually operated ferries. The models differ because the working laws affect the crews' working capacities in the latter case, an absent factor in our setting with autonomous ferries. Keeping all other factors present in both models, a comparison of results allows us to study the differences implied by the working laws prevailing in the manually operated ferries.

* Corresponding author.

E-mail address: mario.guajardo@nhh.no (M. Guajardo).

<https://doi.org/10.1016/j.martra.2024.100115>

Received 24 February 2024; Received in revised form 8 August 2024; Accepted 9 August 2024

Available online 16 August 2024

2666-822X/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

The contribution of this article is twofold. First, our model formulations explicitly capture aspects related to labor laws and crew to different extents. These aspects have not received much attention in the modeling efforts of previous literature debating autonomous versus manually operated ferries. Second, by performing a computational study inspired from a case study in Bergen, a coastline city in Norway, we are able to derive insights on the potential impact of introducing autonomous ferries in a system with other available transportation alternatives. Our results suggest that the fleet of autonomous ferries benefits passengers in terms of less delay times of arrival at their destinations and also in terms of covering more demand. These gains come from the increased flexibility of autonomous ferries with respect to the manually operated fleet, which is more restricted by the working regulations. While the approach of this article is to take a public perspective and minimize passenger delay, the approach can easily be extended to a ferry operator's perspective to understand the impact in revenue from adopting autonomous ferries.

The article is structured as follows. First, Section 2 explores the existing literature. Section 3 presents the problem in detail, together with an integer linear programming model. A computational study and results are presented in Section 4. Section 5 concludes with some final remarks.

2. Literature

A substantial part of the literature on autonomous ships – including ferries – centers around the feasibility, implementation, and assessment of autonomous ships. The literature has focused extensively on Norway as the country leads the technological development of autonomous ships (Munim, 2019). This literature is developing, and several open questions still need to be answered before autonomous ships can be safely used at scale in real-life settings (Gu et al., 2021). Thus far, the current literature has focused on technological advancements, safety considerations, and legal and economic concerns.

2.1. Technological advancements

Most studies on autonomous ships focus on the technological perspective (Munim, 2019), ranging from a project-wide focus to specific problems such as maneuverability, collision avoidance, and docking. Kurt and Aymelek (2022) define the level of autonomy in ships across six levels, from low automation to completely crewless ships, of which most projects to date have yet to achieve the highest level of autonomy.

One project achieving the highest level of autonomy, however, is the “Falco” car- and passenger vessel in the Finnish archipelago—the world's first fully autonomous ferry and began its operations in 2018 (Rolls-Royce Holdings plc, 2018). This project was a milestone in advancing autonomous ferries as most previous attempts at autonomous ships have involved remotely, but still manually, operating the ships. Hence, the project in Finland demonstrated the feasibility of the project considered in Bergen, Norway.

2.2. Safety considerations

Beyond the technological advancements of autonomous systems, a key reason autonomous shipping has gained tremendous momentum in recent years is safety concerns (Xu et al., 2023). While it is believed that human error will not disappear entirely with autonomous ships, 70%–90% of marine accidents today are believed to be caused by such human errors (Porathe et al., 2018). Hence, there is a substantial potential to reap from a safety perspective by transitioning to fully autonomous vessels, and the literature has been heavily focused on this perspective (Gu et al., 2021).

2.3. Legal concerns

When assessing manually operated vessels against autonomous vessels, one essential difference between the two treatments is the working laws protecting the crews. The protection and working limitations of each worker differ because they may be set by multiple parties such as national law and labor unions. However, there are three essential rules to protect most workers—defining how long each work week can be, the minimum amount of rest that must take place between shifts, and the minimum amount of consecutive rest that must take place each week. By Norwegian working laws, a work week can typically be 40 h long, there must be at least 11 h of rest between each shift, and each worker must have at least 35 h of consecutive rest at least once per week (Ministry of Labour and Social Inclusion, 2005).

There are two common approaches to scheduling the weekly rest. A frequently used, albeit simplified, alternative to scheduling these resting days is to schedule five-day work weeks and two consecutive days off. By comparison, a more elaborate approach is to define which shifts each worker should be assigned to and then identify the workers' resting times given the abovementioned requirements (Van den Bergh et al., 2013).

Most personnel scheduling literature covered by the literature review of Van den Bergh et al. (2013) has the following characteristics. First, it usually studies full-time contracts. Second, the shift sequence is typically decided by a scheduling model, in contrast to the assignment of tasks, groups, or time. Third, shifts are typically allowed to overlap. These three characteristics are also followed in this article. Moreover, a slight majority of the personnel scheduling literature has a fixed shift length, while a slight minority has a definable one. The heuristic in this article has the former characteristic, while the exact models have the latter.

2.4. Economical concerns

Economic aspects have commonly played an important role in route planning of vessels, as demonstrated in several recent works that minimize the cost of introducing new energy sources in maritime transportation of passengers (Havre et al., 2022, 2024; Rødseth

et al., 2023). Likewise, several studies attempt to improve the economic aspects of autonomous vessels. These studies typically involve different perspectives on operating costs and capital costs. For instance, employing an operational perspective, Zhang and Wang (2020) attempt to minimize fuel cost consumption and delay penalties of autonomous vessels by assigning berths at each port and deciding the vessel's arrival time at each port. Aslaksen et al. (2021) study the effects of combining fixed schedules and on-demand services provided by a fleet of autonomous ferries in Germany. Employing both an operational cost and a capital cost perspective, Gu and Wallace (2021) address a problem involving decisions on facility location, fleet allocation, and routing in the operations of a water-taxi case in Norway, while Akbar et al. (2021) schedule autonomous vessels to identify the optimal fleet of vessels – in terms of number and size – in addition to the route of each vessel from Europe to Norway and vice versa. Employing a capital cost perspective, Reddy et al. (2019) make the case that the alternatives to zero-emission, autonomous passenger ferries are bridges and manned ferries and that the former is more environmentally friendly, flexible, and cost-effective.

Moreover, a frequent economic argument for using autonomous ships is that *personnel costs* will be reduced. This argument is discredited by Kooij and Hekkenberg (2021), who study how autonomous systems affect crew sizes and argue that autonomous systems will reduce the need for various crew members for *phases* of the trip – such as the sailing, arrival and departure, and loading and unloading phases – and not the entire trip itself. Until full autonomy is achieved, the authors demonstrate that introducing autonomous systems may *increase* the total costs as autonomous systems face operating and maintenance costs. Since ships that have not yet reached full autonomy will not necessarily be able to reduce the number of staff as each role could be demanded for at least one phase of the sailing, the total costs may increase.

Even if full autonomy is achieved, total costs – excluding cost savings from fewer human-made marine accidents – may still not be reduced. Specifically, Kretschmann et al. (2017) studied the cost benefits of autonomous vessels of an autonomous vessel project. The authors found that the reduced crew wages alone are likely not enough to offset the increased costs of fully autonomous vessels—such as increased shore control center costs, increased demand for maintenance crews, and increased capital costs for autonomous ship technology.

While the previous studies on reducing the number of personnel do not indicate a reduction of total costs from reducing crews, even with fully autonomous systems, little has been said on whether fully autonomous systems have the potential to improve *revenues*. Specifically, the crews' working times are limited by working laws, whereas fully autonomous systems are not. In a practical setting, this could imply that fully autonomous systems, such as autonomous ferries, could operate at all hours of the day except for when maintenance is needed. By comparison, manually operated ferries would also be limited by the number of crews available to work within the constraints of the working laws, which could cause a poorer utilization of the fleet of ferries and ultimately lead to fewer passengers being transported than autonomous ferries. Hence, there may be an increased revenue potential from using a fully autonomous fleet.

The revenue potential can be identified by studying changes in the volume of passengers. Specifically, as revenue depends on both pricing and quantity, and deciding prices is a separate decision that affects passenger demand, it is valuable to isolate the problem by assuming similar prices for autonomous and manually operated ferries. Hence, the revenue problem is reduced to a passenger quantity question, which we attempt to explore in this article by studying how much better autonomous ferries may meet the passenger demand throughout a whole week compared to manually operated vessels. As far as we know, this research question has yet to be explored in the literature.

3. Problem and model formulation

With the changing regulations allowing fully autonomous vessels in national and international waters, understanding the impact on passengers of using autonomous ferries instead of manually operated ferries is highly topical. For this purpose, in this section, we introduce a problem with and without the limiting aspects on crews working in manually operated ferries. We organize the section in two parts. The first one provides a verbal description of the problem, and the second one provides its mathematical formulation.

3.1. Problem description

Given a fleet of ferries, a set of time slots spanning a one-week time horizon, and a set of passengers who need to travel within a network of locations, our problem consists of creating routes and schedules for the ferries to perform the transportation as to minimize unmet demand and passenger delays. To capture the demand from passengers, we introduce the concept of a “passenger group”—a group of passengers of various sizes sharing the exact origin, destination, preferred departure time, and maximum willingness to wait. The purpose of passenger groups is to create a realistic variation in the demand for ferries throughout the week-long schedule while reducing the complexity of solving the problem, as whole passenger groups can be allocated instead of each passenger. The ferries are assumed to have no capacity constraints, so the models will either allocate a ferry to transport all or none of the passengers in the passenger group. Passenger groups will either use ferries as their mode of transportation or use alternative transportation. The passengers can only be assigned to a ferry if a relevant route has been assigned during the time window in which passenger groups are willing to wait for the ferry. If no such ferry has been allocated during this time window, the passenger group will use alternative transportation (e.g. a bus), which for purposes of the fleet of ferries counts as unmet demand.

The ferries and passengers move within a network of locations or *stops*. For ferries, it is important to ensure that they are defined logically, each being maximally at one location at a time and arriving from the location it arrived at in the previous time slot. The ferries must either be operative or on stand-by at a stop, of which the latter is defined as allocating the ferry to a two-stop route of one time slot's duration, with the origin and the destination being the exact location. These routes in which a ferry is on stand-by are

called “trivial routes”, and these routes are beneficial if waiting for a future passenger group generates a higher objective function value than starting another route or transporting a passenger group at the present time.

With the general setting above applying to both autonomous and manually operated ferries, other aspects vary. First of all, in contrast to the case of autonomous ferries, when the fleet is operated manually the number of ferries that can be used simultaneously is limited by the number of crews available. Crews also face working restrictions. Specifically, crews cannot work longer than eight-hour workdays, must rest for at least 11 h after each shift, and must have at least one 35-h block of continuous rest each week. Additionally, when exploring to which extent autonomous ferries may meet passenger demand better than manually operated ferries, the following assumptions are made about ferries and crews. First, we assume the fleet of ferries is homogeneous, with the same transportation time for autonomous and manually operated ferries, and without capacity limitations or maintenance needs. Second, we assume homogeneous crews and that the same crew members always work together so that we can treat them as one joint entity of workers. Additionally, we assume that no overtime is allowed for all crews, shift changes between crews are instantaneous, and the ferry does not need to reach a particular stop to perform the shift change.

Assumptions are also made for schedules, docks, and passengers. First, we assume no fixed patterns are necessary for the ferry schedule, such as line 1 running every 30 min daily during daytime hours. The transportation time between two stops is deterministic and equally long for manually and autonomously operated ferries. Second, we assume unlimited docking capacity, meaning there will be no congestion when multiple ferries simultaneously arrive at the same stop. Third, we assume that no passengers want to start and end at the same stop, that they will travel with one ferry only to reach their end destination, and that they are willing to be delayed to their arrival destination for a limited amount of time when using the ferry—otherwise, they will use alternative transportation methods such as a bus.

We formulate two models with the same objective function to compare schedules with crewed and autonomous ferries. Since the ferries are supposed to be a solution to offload to the existing transportation network, two considerations must be made in the objective function in both models. First, to attempt to transport as many passengers as possible who prefer a ferry, while the second consideration is to minimize the arrival delay the transported passengers face. By employing the same objective function and a subset of constraints on the same test instances, we can isolate the effect autonomous ferries have compared to the manually operated ones.

It is important to remark that, besides the passengers’ perspectives adopted in this problem, other criteria regarding safety and economics concerns are also part of the debate on autonomous versus manually operated ferries, as discussed in our literature review. However, to keep the results of the two models comparable and to assess the differences implied by the working regulations only, we do not consider safety or economic criteria. Instead, we keep all aspects equal in both models, except for the different working restrictions commented above.

3.2. Integer linear programming formulation

In the following, we develop a mathematical formulation of the problem, using a framework of integer linear programming. We formulate one model for the fleet of autonomous ferries and another one for the fleet of manually operated ferries. As many of their components are the same, we structure their presentation jointly and classified by group of related constraints. In the formulation of these constraints, we describe how they are tailored for one fleet or another one. Table 1 defines some notation on sets, parameters and decision variables.

3.2.1. Passenger constraints

$$y_p^A + \sum_{\substack{(r,t) \in P_p: \\ (p,r,t) \in \mathcal{O}}} y_{prt} = 1 \quad \forall p \in P \quad (1)$$

$$y_{prt} \leq \sum_{v \in V} x_{vrt} \quad \forall (p, r, t) \in \mathcal{O} \quad (2)$$

Constraints (1) state that each passenger group either goes on a specific route or uses an alternative transportation method like a bus. Alternative transportation must be allocated if the passenger group is not picked up during the *departure window*, consisting of the preferred departure time and the three time slots ahead.

Moreover, passengers can only be assigned if a ferry arrives for relevant routes and time slots. This condition is captured by constraints (2). In this context, the relevant routes are the routes that contain both the origin and destination locations of the passenger group, with the origin being visited before the destination on the route. The relevant time slots are the starting time for the route that makes the ferry arrive at the origin location within the passenger group’s departure window.

3.2.2. Working requirements constraints

Trivial routes have one primary function for autonomous ferries and two functions for manually operated ferries. For both types of operations, it may be beneficial for the objective function if a ferry does not transport passengers at a particular time slot but instead waits at a specific location before departing at a later time slot—for example, because a large number of passengers would like to depart only a couple of time slots later. Hence, it is important not only to capture allocations of ferries in use but also their

Table 1
Model components.

Sets	
T	A circular set of time slots.
R	Set of candidate ferry routes—see Section 4.1 for more details.
V	Set of ferries in the autonomous operations model and the set of crews in the manual operations model.
P	Set of passenger groups.
S	Set of ferry stops.
Subsets	
$\mathcal{P}_p \subset (R, T)$	Set of route–departure time alternatives for passenger group p .
$\mathcal{R}_t \subset (R, T)$	Set of ongoing routes at time t .
$\mathcal{V} \subset R$	Set of trivial, vacant routes staying still on a particular stop for one time slot.
$\mathcal{S}_{st} \subset (R, T)$	Set of routes starting at stop s at time t .
$\mathcal{E}_{st} \subset (R, T)$	Set of routes ending at stop s at time t .
$\mathcal{O} \subset (P, R, T)$	Tuple set of (passenger, route, departure time) options.
$\mathcal{H}_t^{11} \subset T$	Set of 11-h contiguous time slots starting at time t .
$\mathcal{H}_t^{19} \subset T$	Set of 19-h contiguous time slots starting at time t .
$\mathcal{H}_t^{35} \subset T$	Set of 35-h contiguous time slots starting at time t .
Parameters	
$\delta_{prt} \geq 0$	The arrival delay that passenger group p faces by taking route r that begins at time t .
$\rho_p \geq 0$	Number of passengers in passenger group p .
$\alpha_p > \max_{p,r,t} \delta_{prt}$	Penalty if passenger group p must use alternative transportation.
$\nu \geq 0$	Max number of ferries operating simultaneously.
$\eta \geq 0$	Max time slot per week the ferry/crew can work.
$\mu_v \geq 0$	Min time slots per week that ferry/crew v must work.
$\theta_r \geq 0$	Duration of route r , as measured in time slots.
Variables	
$x_{vrt} \in \{0, 1\}$	Ferry v starting route r on time t .
$y_{prt} \in \{0, 1\}$	Passenger group p is assigned to the route r that starts its route at time t .
$y_p^A \in \{0, 1\}$	Passenger group p is assigned to alternative transportation.
$z_{vt}^R \in \{0, 1\}$	Ferry/crew v rests on time t .
$z_{vt}^D \in \{0, 1\}$	Ferry/crew begins its daily rest on time t .
$z_{vt}^W \in \{0, 1\}$	Ferry/crew begins its weekly rest on time t .

geographical position when they are not in use so that they depart from this position next. By assigning the ferry a series of trivial routes when the ferry is not operational, this benefit can be captured.

$$\sum_{r \in \mathcal{V}} x_{vrt} + \frac{1}{11} \sum_{\tau \in \mathcal{H}_t^{11} \text{ prev}(t, 10)} z_{v\tau}^D \geq \left(1 + \frac{1}{11}\right) z_{vt}^R \quad \forall v \in V, t \in T \quad (3)$$

Additionally, manually operated ferries use trivial routes to define whether a particular crew is resting, a critical concern when employing working requirements to the model. Hence, we define z_{vt}^R to equal 1 when a particular crew and their ferry v rests – i.e., not working – on time t . This rest is defined to occur when two criteria are met. First, the crew's ferry is assigned a trivial route at time t , meaning no passengers are transported, and the ferry is on standby. Second, the crew is undertaking their daily rest of minimally 11 h at time t . Constraints (3) define when z_{vt}^R is equal to zero.

$$\sum_{r \in R} \sum_{t \in T} \theta_r x_{vrt} - \sum_{t \in T} z_{vt}^R \leq \eta \quad \forall v \in V \quad (4)$$

$$\sum_{r \in R} \sum_{t \in T} \theta_r x_{vrt} - \sum_{t \in T} z_{vt}^R \geq \mu_v \quad \forall v \in V \quad (5)$$

Furthermore, the x variable denotes a ferry's location and route. While it does not denote if the ferry is actively being used, we do know the hours the crew rests. Hence, we can deduct those hours to get the actual number of hours worked to enforce that each crew can maximally work a $\eta = 40$ hour work-week, which constraints (4) state. By a similar approach, we can enforce a minimum work-week duration per crew of μ_v , which constraints (5) state.

$$\sum_{\tau \in \mathcal{H}_t^{11}} \sum_{(r,t) \in \mathcal{R}_\tau: r \notin \mathcal{V}} x_{vrt} \leq |\mathcal{H}_t^{11}| (1 - z_{vt}^D) \quad \forall v \in V, t \in T \quad (6)$$

$$\sum_{\tau \in \mathcal{H}_t^{19}} z_{v\tau}^D \geq 1 \quad \forall v \in V, t \in T \quad (7)$$

$$\sum_{\tau \in \mathcal{H}_t^{35}} \sum_{\substack{(r,\tau) \in \mathcal{R}_\tau: \\ r \notin V}} x_{vr\tau} \leq |\mathcal{H}_t^{35}|(1 - z_{vt}^W) \quad \forall v \in V, t \in T \quad (8)$$

$$\sum_{t \in T} z_{vt}^W \geq 1 \quad \forall v \in V \quad (9)$$

Moreover, working requirements motivate the formulation of constraints concerned with resting times. First, we must enforce that the daily rest is at least 11 h long and how often they should minimally occur. When enforcing the former, it is sufficient to define when the daily rest does not occur—which constraints (6) state is the case if no ongoing routes have been assigned to crew v on time t . Additionally, we must ensure that the daily rest occurs minimally every $11 + 8 = 19$ hours since workdays cannot be longer than 8 h long, and constraints (7) ensure this. Second, we must enforce that at least one 35-h period of continuous rest occurs at least once during the week, which requires two constraints. Similarly to the daily rest, it is sufficient to define when the weekly rest does not occur, and constraints (8) follow a similar approach for the weekly rest as constraints (6) do for the daily rest. Finally, constraints (9) ensure that a minimum of one such 35-h resting period must be enforced throughout the week.

3.2.3. Ferry constraints

$$\sum_{(r,\tau) \in \mathcal{R}_t} x_{vr\tau} \leq 1 \quad \forall v \in V, t \in T \quad (10)$$

$$\sum_{(r,\tau) \in \mathcal{E}_{s,prev(t)}} x_{vr\tau} = \sum_{(r,\tau) \in \mathcal{S}_{s,t}} x_{vr\tau} \quad \forall v \in V, s \in S, t \in T \quad (11)$$

$$\sum_{v \in V} \sum_{(r,\tau) \in \mathcal{R}_t} x_{vr\tau} \leq \nu \quad \forall t \in T \quad (12)$$

Beyond the passenger and working requirements constraints, three more groups of constraints are needed to target the ferry logic specifically. That is, a ferry can maximally be at one route on each time slot, and ferries can only begin a route at a stop they arrived at in the previous time slot. Constraints (10) and (11) ensure these conditions, respectively.

Moreover, constraints (12) state that a maximum capacity of simultaneously operating ferries must be respected. While this condition is inherently respected by the set V in the model for autonomous ferries, we incorporate it in the formulation of constraints in the model for manually operated ferries to ensure that the number of ferries employed simultaneously does not surpass the number of crews.

3.2.4. Variable constraints

The model concludes with constraints on the binary domain of the variables below.

$$x_{vrt} \in \{0, 1\} \quad \forall v \in V, r \in R, t \in T \quad (13)$$

$$y_{prt} \in \{0, 1\} \quad \forall p \in P, r \in R, t \in T \quad (14)$$

$$y_p^A \in \{0, 1\} \quad \forall p \in P \quad (15)$$

$$z_{vt}^R \in \{0, 1\} \quad \forall v \in V, t \in T \quad (16)$$

$$z_{vt}^D \in \{0, 1\} \quad \forall v \in V, t \in T \quad (17)$$

$$z_{vt}^W \in \{0, 1\} \quad \forall v \in V, t \in T \quad (18)$$

3.2.5. Objective function

The objective function is to minimize the total penalty occurring from the number of passengers ρ in passenger group p , the delay penalty relative to the earliest possible arrival δ_{prt} for passenger group p , and the penalty α_p occurring because the passenger group chose an alternative transportation method.

$$\min_{y, y^A} \sum_{p \in P} \rho_p \left(\sum_{(r,t) \in \mathcal{P}_p} \delta_{prt} y_{prt} + \alpha_p y_p^A \right) \quad (19)$$

While the integer linear programming formulation covers the entire problem under the assumptions mentioned earlier, a test instance with many passenger groups, in particular, may render the model computationally intractable. Specifically, the biggest driver for the number of decision variables in a test instance is the number of passenger groups, which depends on the number of departure stops, arrival stops, the number of time slots in the schedule, and how many of these (departure stop, arrival stop, time

Table 2

Travel time between each stop, measured in time slots (each time slot lasts five minutes).

	1	2	3	4	5	6
1	0	1	2	3	2	1
2	1	0	1	2	3	2
3	2	1	0	1	2	2
4	3	2	1	0	1	2
5	2	3	2	1	0	1
6	1	2	2	2	1	0

slot) permutations have a non-zero demand. The duration of a time slot should preferably be short – such as five minutes – and there should preferably be a demand for transportation in many of the time slots to create realistic test instances. Moreover, the entire schedule should be at least one week long when considering labor regulations to capture the most important constraints, such as daily and weekly rest. With a week-long schedule consisting of short time slots with a non-zero demand, the number of passenger groups will be high, eventually making the exact model time-consuming or even computationally intractable.

Consequently, we have developed a heuristic approach, which is described in the [Appendix](#). A heuristic method to address the problem is valuable as it balances the trade-off between solution quality and computational efficiency. While adhering to the same assumptions as the exact model, our heuristic introduces an additional assumption regarding crew scheduling—that crews are assumed to work eight-hour shifts, commencing simultaneously each day for five consecutive days. This simplification will likely result in a lower solution quality than the exact model for realistic test instance sizes. However, it remains a reasonable assumption in light of the Norwegian Working Environment Act, which emphasizes consistent working hours and mandates a continuous weekly rest period for employees.

4. Computational study

Similar to many urban areas, Bergen, Norway faces transportation challenges. With a growing population, a surge in tourism, and public transportation that resembles a hub-and-spoke structure – with most buses going via the city center due to the coastline and mountainous surroundings – the city's transportation system is congested and under pressure to accommodate the increasing demand for mobility.

This challenge spurred the interest of the “blue light rail” project ([Opus, 2024](#)). The project aims to provide an alternative mode of transportation for the city's residents and tourists to reduce traffic congestion and improve the overall transportation experience. The concept of the project is still in the planning phase, and further research is needed to evaluate its feasibility. However, despite being in the planning phase, the project has gained significant interest because the fjords are an integral part of the city of Bergen and play a significant role in shaping its history, culture, economy, and political identity. The current suggestion involves six stops for the blue light rail, and [Table 2](#) shows the hypothetical travel times between each of these stops, as measured in time slots used in the models presented in this article. The length of a time slot is five minutes.

4.1. Generating passenger groups and routes

The popularity of a passenger group was calculated as follows. First, each station, weekday, and hour are given popularity scores, with weekends and nights being notably less popular than other weekdays and hours. Second, static popularity scores of the origin and destination locations, weekday, and hour are multiplied. A popularity parameter is then calculated

as $pop_{pg} := \ln(1 + \prod_{i=1}^4 p_{i,pg}) \forall pg$, in which p is a vector that represents each of these four metrics and pg represents the passenger group. Third, the popularity score determines the number of passengers in a passenger group, which is calculated as $\lceil (pop_{pg} - \min_j pop_j)^{(1/\beta)} \rceil \forall pg, \beta > 0$. Fourth, only n of these passenger groups are kept for the test instance, in which the probability of being drawn is $\frac{pop_{pg}}{\sum_j pop_j} \forall pg$.

Realistic routes were created and added to the set of candidate routes. In this context, a realistic route is defined as a route with a low total overall travel distance compared to other route alternatives involving the same locations. When creating this set, all two-stop routes were generated, including the six “stand-by routes”—one for each stop. In contrast, the majority of the 1200 permutations for routes of length three to five stops are discarded because they involve larger distances traveled than alternative routes involving the same stops. Specifically, only 100 of these routes satisfy the criteria for being considered realistic, where 76 are of length three, 12 of length four, and 12 of length five stops. For routes of length six, each stop is the origin location for two routes—one going clockwise and the other going counter-clockwise, visiting all stops. Routes of length seven are the same as the routes of length six, except that the origin location is appended as the destination location. [Table 3](#) shows the number of candidate routes selected per each of the seven possible lengths.

Table 3
The number of routes per given length.

Stops	2	3	4	5	6	7
Number of routes	36	76	12	12	12	12

Table 4
Results with the exact models.

Instance	1	2	3			
Passenger groups	200	200	10000			
Time slots	168	2016	2016			
Ferries	1	15	15			
Crews	1	15	15			
Objective function value, no assignments	10554	10512	524448			
Treatment	A	M	A	M	A	M
Heuristic						
Objective function value	6730	9016	517	694	108587	234658
Exact model						
Objective function value	5553	8333	0	NA	235642	NA
Relative MIP gap (percent)	0	0	0	NA	100	NA
Variables	43241	43745	4854850	4945568	5658150	5748868
Constraints	17537	18380	228128	351149	1031430	1154449
Running time (seconds)	2.4	9263	2839	NA	21600	NA

4.2. Smaller test instances

Computations were undertaken on a Linux Ubuntu 18.04 computer with an 18-core, hyper-threaded 2.6 GHz Intel Xeon Platinum 8272CL processor and 144 GB RAM. The exact models' test instances were run with the mathematical programming language AMPL and the commercial solver CPLEX version 12.10.0 – of which CPLEX ran each instance for six hours unless optimality was found earlier – and the heuristic was run in a parallelized mode of R version 3.6.3.

Table 4 highlights the challenges of the exact models. Specifically, test instance one is a small problem consisting of 200 passenger groups to be assigned, one-hour long time slots, one ferry and one crew. This test instance solves to optimality for autonomous and manually operated ferries, but the latter problem variant takes over 2.5 h to solve to optimality.

Test instance two further highlights the challenges for this manually operated test variant. While the number of passenger groups is still 200, the number of time slots has increased to five-minute time slots, and both the number of ferries and crews has increased to 15. This change increases the number of decision variables 113-fold and the number of constraints 19-fold. **At this problem size, CPLEX cannot run the problem variant. By comparison, the autonomously operated test variant can still solve the problem to optimality, but the solving time increased from 2.4 s to 47 min between the two test instances.**

However, the autonomously operated problem variant faces challenges when the number of passenger groups increases from 200 to 10000. While 6003 passenger groups are allocated to ferries, the MIP gap is still 100%, and the objective function value is dramatically improved in the heuristic. Hence, the exact models produce poor results at realistically large problem sizes.

The results above suggest that the exact models are inefficient to address larger instances, where even finding a feasible solution might become challenging. In contrast, the heuristic allows to find feasible solutions to all instances in reasonable time, although at sacrifice of solution quality.

4.3. Larger test instances

We now aim at exploring larger size instances, for which we adopt the heuristic instead of the exact approach. This section studies 300 test instances with 5-minute time slots and 10000 passenger groups demanding transportation. Such small time slots and such a high number of passenger groups result in a more realistic representation of the real-life demand of ferries, and these time slot sizes and passenger group numbers result in an average of 4.96 passenger groups per time slot demanding transportation.

The test instances differ in which passenger groups are present in the set of passenger groups, P . With 5-minute time slots and six locations, the 10000 passenger groups are drawn from a set of 60480 alternatives. These passengers groups are drawn randomly, as described in Section 4.1.

4.3.1. Description of test instances

Figs. 1 and 2 show the distribution of passengers and groups. With six origins, five destinations, and five-minute time slots, there are 60480 (origin, destination, departure time) permutations during a week. 10000 passenger groups are then drawn and given a non-zero demand, both of which are drawn based on the group's "popularity"—which is the demand of leaving a particular origin at a specific time slot to go to a particular destination. The number of passengers within these groups is illustrated for test instance 1 in Fig. 1, with a mean of 8.6 passengers per passenger group and a standard deviation of 2.6 passengers. The most frequent passenger

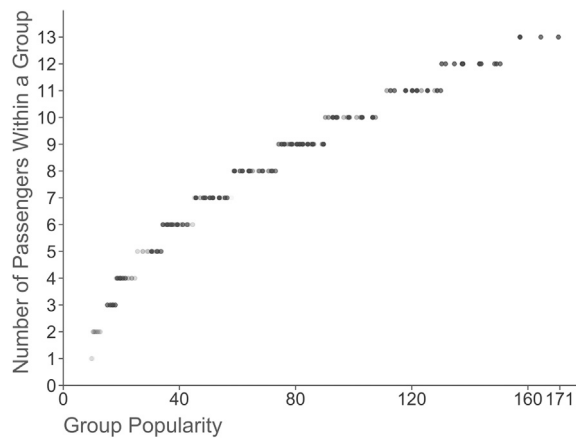


Fig. 1. Number of passengers given the passenger group's popularity.

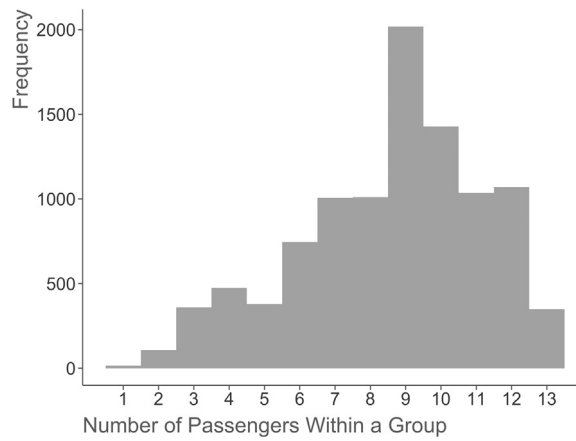


Fig. 2. Distribution of passenger group sizes.

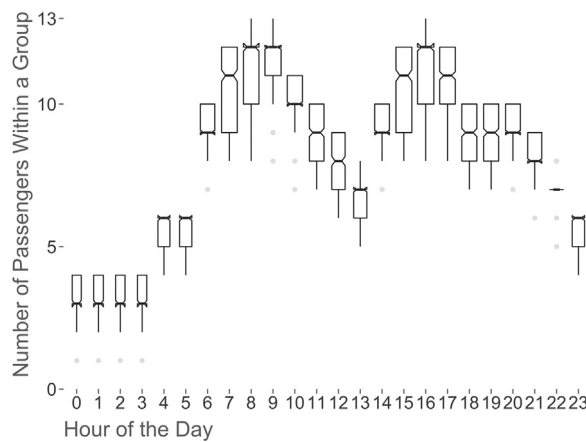


Fig. 3. Distribution of passenger group sizes given preferred departure hour.

group size consists of nine passengers, and the majority of the passenger groups have between nine and 12 passengers. Throughout the week, 85920 passengers will be transported in test instance 1 if all passenger groups are allocated.

The number of passenger groups and passengers also varies throughout the day and week. Specifically, Figs. 3 and 4 show that the number of passengers within each passenger group is higher Monday–Friday and in the typical rush hours. Additionally, the

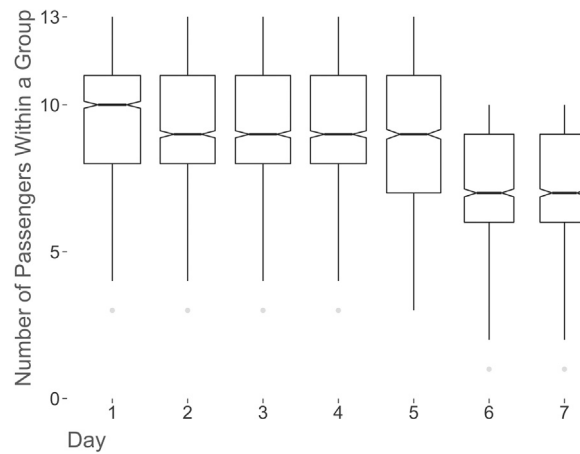


Fig. 4. Distribution of passenger group sizes given preferred departure day.

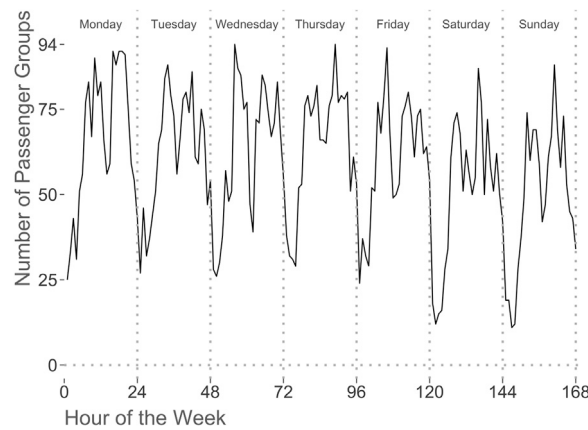


Fig. 5. Number of passenger groups per hour of the week.

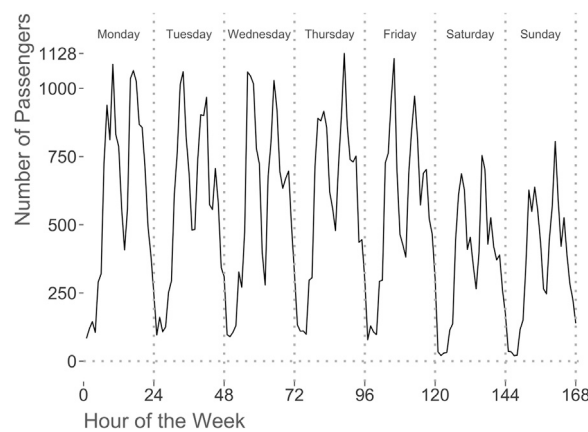


Fig. 6. Number of passengers per hour of the week.

number of passenger groups fluctuates during the day and week as well, which Fig. 5 illustrates. Consequently, as the number of passengers is the sum product of the number of passenger groups and the number of passengers within each group, the number of passengers demanding transportation throughout the week fluctuates with rush hours and weekdays—as can be seen in Fig. 6.

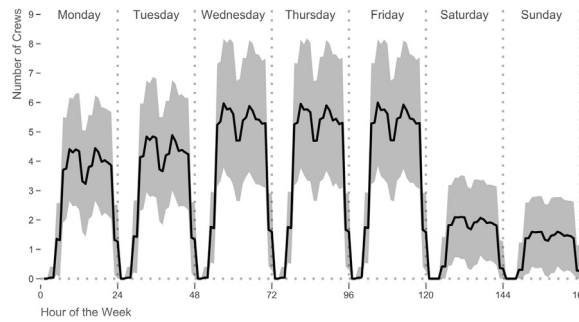


Fig. 7. The highest number of crews working simultaneously for each hour of the week. The black line represents the average value across test instances while the shaded area represents the 95% confidence interval.

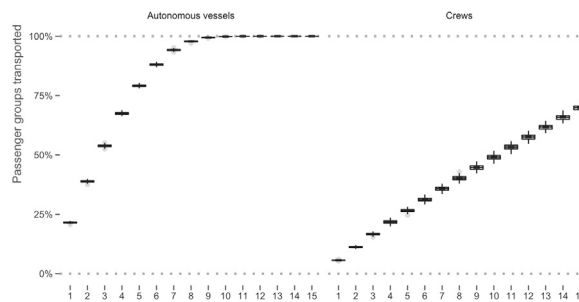


Fig. 8. Passenger demand met by ferry fleet size, as measured in the ratio of passenger groups being allocated.

4.3.2. Description of solution

Since the models' objective is to minimize the penalties function for unmet demand and delays, it is natural that many crews are assigned to weekdays, especially at peak hours when the demand tends to be higher. As opposite, less crews are assigned to late night hours and weekend hours, where demand is lower. This effect is demonstrated by Fig. 7, in which none of the 15 crews are scheduled for the first hours of each day, and most crews are allocated for the peak hours during the day—with the most crews being allocated in the morning and afternoon. We remark here that

Consequently, many simultaneously working crews create an extensive fleet of ferries used only a few hours during the day. For instance, when 15 crews were working, the highest number of simultaneously working crews across all test instances was 8. These under-utilization tendencies exist across all 300 test instances. Hence, the following calculation was made for each test instance to fairly compare autonomously and manually operated ferries for a given fleet size. First, for each crew 1 through 15, we identified how many crews worked on each time slot. Second, we identified the highest number of simultaneously working crews for each crew, equivalent to the lowest number of ferries needed for the schedule. This lowest number of manually operated ferries is the number we compare against an equally sized fleet of autonomous ferries within each test instance.

The low fleet utilization naturally manifests itself in meeting less demand, too. Fig. 8 shows a near-linear relationship between the number of crews transporting passengers and the ratio of passengers being transported, indicating that even with a two-digit number of crews, there are still substantial gaps in the schedule with plenty of unmet demand for ferry transportation. In comparison, autonomous ferries reach a median coverage of 100% across test instances with 12 autonomous ferries and 97.8% with eight ferries. While the percentage difference does decrease between the two types of ferries as the autonomous ferries converge towards 100% coverage, autonomous ferries still cover minimally 43.17% more passenger groups than manually operated ferries for any number of ferries/crews below 15.

Each hour of the day was assigned one out of eight ordinal scores of how many passengers could be expected to demand a ferry ride, all else equal. Generally, the score is lower in the night and higher during the day, with the lowest scores at 00:00–03:59 and the highest scores at hours 08:00–09:59 and 16:00–16:59. As can be seen from Fig. 9, autonomous ferries transport more passengers per time slot across all hour popularities than manually operated ferries. This difference increases the greater the hour popularity and fleet size.

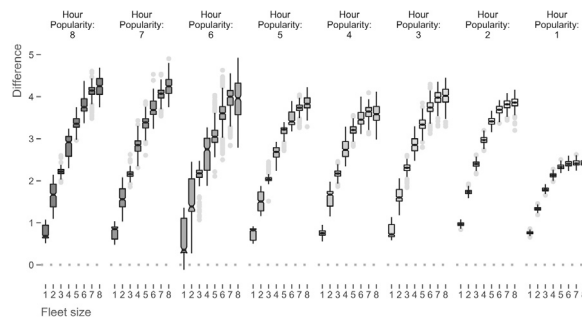


Fig. 9. Surplus coverage – as measured in number of passengers – by autonomous ferries per time slot compared to manually operated ferries.

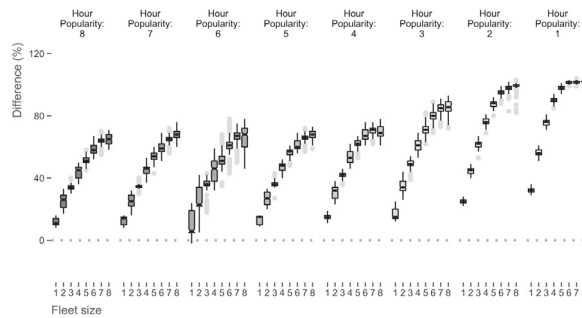


Fig. 10. Percentage-wise surplus coverage by autonomous ferries per time slot compared to manually operated ferries.

While Fig. 9 indicates that autonomous ferries can contribute the most value if introduced at popular hours, Fig. 10 suggests the opposite. Fig. 10 shows, percentage-wise, how many more passengers the autonomous ferries transport compared to manually operated ferries for a given fleet size and hour popularity. Naturally, popular hours include more passengers demanding transportation, which is a driver for why an outperforming transportation method – in this case, autonomous ferries – will transport more passengers. If one transforms the outperformance to a percentage of total demanded transportation for the time slot in question, Fig. 10 shows that autonomous ferries provide the greatest coverage for the least popular hours of the week. This outperformance is caused by the fact that autonomous ferries could cover all demand with a sufficient number of ferries. In contrast, manually operated ferries prioritized the more popular hours of the week. Considering that it may not be possible to operate autonomous ferries all hours of the week in reality and that there are other, manually operated transportation alternatives available – such as buses – which also tend to prioritize popular hours of the week, this finding indicates that autonomous ferries may provide great value when used in combination with other transportation alternatives if used during low-demand hours of the day.

Figs. 11 and 12 strengthen the hypothesis of providing autonomous ferries at times of the week with less demand. Specifically, each day of the week was assigned an ordinal score, with Monday–Friday ranked equally and Saturday–Sunday ranked lower. Just as with hours of the day, autonomous ferries outperform manually operated ferries the most in the percentage of the time slot's total demand when there is lower demand—which is during the weekend from a day-wise perspective. However, autonomous ferries also outperform manually operated ferries the most in terms of the number of excess passengers they provide *despite* the fact that the weekend has less demand for passengers. This outperformance contrasts the hourly perspective, in which Fig. 9 showed a lower outperformance in terms of excess passengers transported for the least demanded hours of the day. Consequently, if few autonomous ferries are available with limited operating time per week, employing autonomous ferries during the weekends may be beneficial when fewer manually operated alternatives are available.

5. Concluding remarks

This article assessed the flexibility of autonomous ferries in transport operations subject to labor regulations, motivated by the case of Bergen, a coastal city in Norway where water buses are currently under consideration. To assess this problem, we

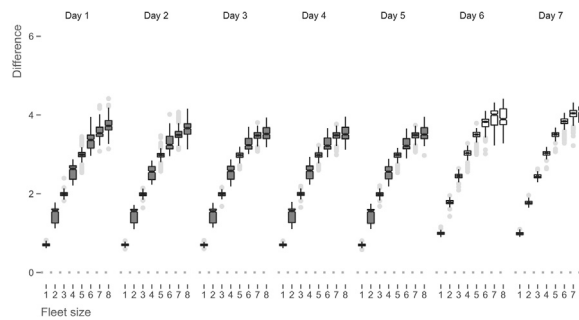


Fig. 11. Surplus coverage—as measured in number of passengers—by autonomous vehicles compared to manually operated ferries across all ferry fleet sizes.

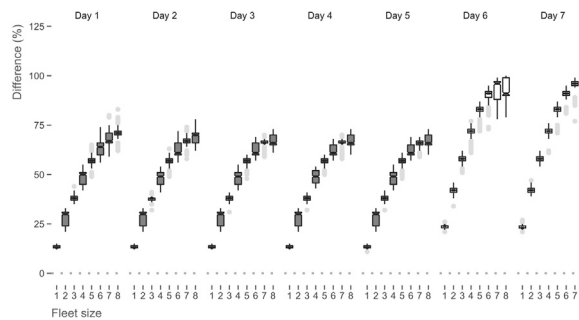


Fig. 12. Surplus coverage by autonomous vehicles, as calculated on an adjusted, *percentage-wise* time slot-basis across all ferry fleet sizes.

formulated two integer linear programming models whose objective function minimizes the total penalty due to the arrival delay from transported passenger groups, the penalty of passenger groups that use alternative transportation, and the number of passengers within each group.

Our computational study showed that the majority of the manually operated ferries are allocated to work during peak hours. Since crews can only work for eight hours before needing a daily rest, these simultaneous work allocations cause a low ferry utilization as the ferries are only operated for some of the day. Consequently, autonomous ferries transported more passenger groups throughout the day than their manually operated counterparts, and this effect happened for all fleet sizes. Moreover, employing only a handful of autonomous ferries that operate 24/7 to cover most of the transportation demand was sufficient.

While fleet utilization could be improved by hiring more crews to operate on time of the week with less demand, it may make little economic sense to do so. Hence, this outperformance by autonomous ferries will likely remain substantial in a practical setting, too. For the test instances presented in this article, autonomous ferries outperformed manually operated ferries throughout the week. However, the severity of it depended on the hour and day of the week. Specifically, autonomous ferries transport more passengers per time slot than manually operated ferries for higher day and hour popularities and bigger fleet sizes.

However, when measuring the outperformance in the percentage of total demand for each time slot, autonomous ferries outperformed the manually operated ferries the most at low-demand hours—at hours and days of the week with the lowest popularity scores. Due to potential practical limitations in the number of operating hours for autonomous ferries and due to other transportation alternatives that will operate the most during high-demand hours, this finding indicates that autonomous ferries may provide significant value when used in combination with other transportation alternatives if used during low-demand hours of the day. Thus, we recommend autonomous ferries to be used from late evenings to early mornings in addition to weekends.

Although our work has focused on passengers' perspectives, the results hold significant for policymakers and transport company managers. In fact, public transport authorities would normally advocate for transport systems that can meet population's needs. Our results have shown that using autonomous ferries in public transportation can significantly improve the passengers' utility, particularly in Norway, where there is a high coast-to-land ratio, road infrastructure is expensive, and a positive outlook on AVs by authorities makes this technology particularly relevant. Moreover, the national debate points to the autonomous ferries technologies coming along with electric batteries instead of fossil fuels, which from environmental policy perspective is also an important aspect. From a managerial perspective, our results have shed light on how to allocate transport fleet resources across a day as to meet common demand patterns with autonomous ferries. In this respect, our results have shown that there may be a substantial potential

for increased revenues by employing autonomous vessels, as the number of passengers transported increased with respect to the manually operated fleets.

As for future research, exploring economic objectives in this problem is important, as replacing a fleet of manual by autonomous ferries will naturally involve an investment cost, which is absent in our paper. Also, as the integer models proved hard to solve, further algorithmic developments to improve the heuristic approach in this paper remain of interest.

CRediT authorship contribution statement

Christian Braathen: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Julio C. Goetz:** Writing – review & editing, Resources, Project administration, Conceptualization. **Mario Guajardo:** Writing – review & editing, Resources, Project administration, Conceptualization.

Declaration of competing interest

There are no financial or non-financial interests that are directly or indirectly related to the work submitted for publication.

Declaration of Generative AI and AI-assisted technologies in the writing process

No generative AI tools have been used for the writing process.

Acknowledgment

We thank Stein W. Wallace and the team of the AUTOLOG project (project number 280536, Research Council of Norway) for exciting discussions about the problem studied in this article.

Appendix. Heuristic

In this section, we describe the core components of our proposed solution approach, which consists of an algorithm to determine the work schedule for a crew and ferry. The primary algorithm – Algorithm 1 – is responsible for constructing the work schedule by evaluating multiple alternatives. The main intention of this algorithm is to allocate passenger groups to fit gaps in the schedule so that the number of allocated passenger groups can be high and their delay and alternative transportation penalties can be low. For an extensive explanation of the heuristic, see [Braathen \(2024\)](#).

The algorithm aims to create a work schedule efficiently for each ferry in a fleet. The algorithm is designed to iteratively evaluate potential workweeks, starting with initializing an empty `WorkSchedule` list that stores the most favorable schedule for each ferry, considering the various constraints and objectives.

For each *ferry*, the algorithm initializes an empty `PotentialWorkWeek` list and assesses various *schedule starting times*—the first day only for the autonomous ferries and each day of the week for crews. This list represents the best-chosen schedule for a given starting block and allows us to compare and select the most suitable workweek allocation.

It then examines each *working day* within a week—seven days ahead of the week for autonomous ferries from the schedule starting time, and five days ahead for crews. Due to the circular nature of the work schedule, the weekdays are treated as a circular set, and the algorithm uses this circular set to generate the time slots that occur in this potential shift.

The *starting row* loop is introduced to account for the fact that, although passenger groups are sorted by their objective function values when scheduled to reach their destination at their preferred arrival times, the overall objective function value of the entire schedule may be lower if the (time, origin, destination) permutation of the highest-ranked passenger group turns out to be an unfavorable starting point for the schedule at large. By exploring different starting points, the heuristic aims to enhance the overall objective function value, resulting in a better schedule with more passenger groups being transported or facing less delay from preferred arrival times. In our experiments, it became clear that any starting rows beyond five did not provide added value.

Furthermore, the algorithm filters passenger groups based on departure and arrival windows. It initializes an empty `local_schedule` list to construct and store feasible work schedules throughout the algorithm's execution and an empty `passenger_groups_allocated` vector to keep track of the passenger groups that have been allocated to ferries in this local schedule, providing a means to assess the overall efficiency and effectiveness of the alternative work schedules that are produced. The algorithm then attempts to allocate unassigned passenger groups to available time slots, considering whether the origin or both the origin and destination need to be allocated. The algorithm employs the supporting functions that handle the allocation process when allocating passenger groups.

After evaluating and comparing the working day schedules for different starting rows, the algorithm selects the best working day schedule and adds it to the list of potential workweeks. Finally, it identifies the best potential workweek for each crew and ferry and accordingly updates the list of allocated passenger groups.

Algorithm 1: Allocate passenger groups to ferries

```

1 Definitions:
   S          set of time slots for the working shift.
   ed         earliest time slot the passenger group can depart from the origin.
   ld         latest time slot the passenger group can depart from the origin.
2  ea         earliest time slot the passenger group can arrive at the destination.
   la         latest time slot the passenger group can arrive at the destination.
   schedule start day 1 for autonomous ferries and day 1,2,...,7 for manually operated ferries.
Input: list of ferries, list of schedule starts, data frame of passenger group data, list of time slots
Output: finalized work schedule for each ferry

3 initialize an empty WorkSchedule;
4 for each ferry do
5   initialize an empty PotentialWorkWeek;
6   for each schedule start do
7     for each working day from the schedule start do
8       initialize an empty WorkingDay;
9       generate time slots for potential shift;
10      for starting row 1.5 of the passenger groups data frame do
11        move rows < starting row to the end;
12        filter to passenger groups in which  $ld \in S \vee ea \in S$ ;
13        initialize an empty local_schedule and a local copy of allocated passenger groups;
14        for each passenger groups not allocated to local_schedule or previous ferries yet do
15          if any open time slots in  $\{ea, \dots, la\} \cap S$  then
16            if the origin location is already assigned to local_schedule in  $\{ed, \dots, ld\} \cap S$  then
17              if it is sufficient to allocate destination only; then
18                attempt to allocate destination only to local_schedule (Function 1);
19              else
20                attempt to allocate both origin and destination to local_schedule (Function 2);
21              end
22            else
23              attempt to allocate both origin and destination to local_schedule (Function 2);
24            end
25          end
26        end
27        evaluate local_schedule and update the WorkingDay list;
28      end
29      find the best-performing local schedules across different starting rows;
30    end
31    add the best-performing local schedule to the list of potential workweek schedules;
32  end
33  find the best potential working week schedule for the ferry, which is the final schedule for the ferry;
34  update the list of already allocated passenger groups;
35 end

```

References

- Akbar, A., Aasen, A.K., Msakni, M.K., Fagerholt, K., Lindstad, E., Meisel, F., 2021. An economic analysis of introducing autonomous ships in a short-sea liner shipping network. *Int. Trans. Oper. Res.* 28, 1740–1764.
- Aslaksen, I.E., Svanberg, E., Fagerholt, K., Johnsen, L.C., Meisel, F., 2021. A combined dial-a-ride and fixed schedule ferry service for coastal cities. *Transp. Res. A: Policy Pract.* 153, 306–325.
- Braathen, C., 2024. Essays on staffing and transportation. (Phd dissertation). NHH Norwegian School of Economics, Helleveien 30 5045Bergen, Norway.
- Gu, Y., Goetz, J.C., Guajardo, M., Wallace, S.W., 2021. Autonomous vessels: state of the art and potential opportunities in logistics. *Int. Trans. Oper. Res.* 28, 1706–1739.
- Gu, Y., Wallace, S.W., 2021. Operational benefits of autonomous vessels in logistics—a case of autonomous water-taxi in Bergen. *Transp. Res. E: Logist. Transp. Rev.* 154, 102456.
- Havre, H.F., Lien, U., Ness, M.M., Fagerholt, K., Rødseth, K.L., 2022. Cost-effective planning and abatement costs of battery electric passenger vessel services. *Transp. Res. D: Transp. Environ.* 113, 103495.
- Havre, H.F., Lien, U., Ness, M.M., Fagerholt, K., Rødseth, K.L., 2024. Network design with route planning for battery electric high-speed passenger vessel services. *European J. Oper. Res.* 315 (1), 102–119.
- Kooij, C., Hekkenberg, R., 2021. The effect of autonomous systems on the crew size of ships—a case study. *Marit. Policy Manag.* 48, 860–876.
- Kretschmann, L., Burmeister, H.C., Jahn, C., 2017. Analyzing the economic benefit of unmanned autonomous ships: An exploratory cost-comparison between an autonomous and a conventional bulk carrier. *Res. Transp. Bus. Manag.* 25, 76–86.
- Kurt, I., Aymelek, M., 2022. Operational and economic advantages of autonomous ships and their perceived impacts on port operations. *Marit. Econ. Logist.* 24, 302–326.
- Ministry of Labour and Social Inclusion, 2005. The working environment act. Online. Available at: <https://lovdata.no/dokument/NL/lov/2005-06-17-62>.
- Munim, Z.H., 2019. Autonomous ships: a review, innovative applications and future maritime business models. *Supply Chain Forum: Int. J.* 266–279.
- Opus, 2024. Blå bybane. <https://opus.no/projects/blabybane/>. Accessed: 2024-01-21.
- Porathe, T., Hoem, Å., Rødseth, Ø.J., Fjortoft, K., Johnsen, S.O., 2018. At least as safe as manned shipping? autonomous shipping, safety and human error. In: *Safety and Reliability—Safe Societies in a Changing World*. In: Proceedings of ESREL 2018, June 17–21, 2018, Trondheim, Norway.
- Reddy, N.P., Zadeh, M.K., Thieme, C.A., Skjetne, R., Sorensen, A.J., Aanonsen, S.A., Breivik, M., Eide, E., 2019. Zero-emission autonomous ferries for urban water transport: Cheaper, cleaner alternative to bridges and manned vessels. *IEEE Electr. Mag.* 7, 32–45.

- Rødseth, K.L., Fagerholt, K., Proost, S., 2023. Optimal planning of an urban ferry service operated with zero emission technology. *Marit. Transp. Res.* 5, 100100.
- Rolls-Royce Holdings plc, 2018. Rolls-royce and finferries demonstrate world's first fully autonomous ferry. <https://www.rolls-royce.com/media/press-releases/2018/03-12-2018-rr-and-finferries-demonstrate-worlds-first-fully-autonomous-ferry.aspx>.
- Van den Bergh, J., Beliën, J., De Bruecker, P., Demeulemeester, E., De Boeck, L., 2013. Personnel scheduling: A literature review. *European J. Oper. Res.* 226, 367–385.
- Xu, H., Moreira, L., Guedes Soares, C., 2023. Maritime autonomous vessels.
- Zhang, W., Wang, S.A., 2020. Autonomous vessel scheduling. *J. Oper. Res. Soc. China* 8, 391–414.